

SAI S SARVADE

Data Engineer | San Jose, CA | (650) 617-5074 | sai.s.sarvade@gmail.com | [linkedin.com/in/saisarvade](https://www.linkedin.com/in/saisarvade) | sarvade.github.io/portfolio

SUMMARY

Data Engineer with 5+ years of experience turning raw logs into canonical datasets and trusted company metrics across TikTok and Delta Air Lines. Currently own 30+ production pipelines processing 20 TB+ daily and the metric definitions 15M+ users act on, with hands-on experience in dbt, Airflow, SQL, Python/PySpark, Spark, and Trino. Built the data integrity tooling my team now runs on, unprompted, because waiting for permission would have meant shipping wrong numbers for another quarter. Partner with product, analytics, and research teams to define metrics, set data standards, and build self-serve data products.

EXPERIENCE

TikTok Inc. | Data Engineer

San Jose, CA | May 2025 to Present

Creator Compass & Seller Compass, the self-serve analytics products 15M+ users rely on across 19 countries

- Define and manage the multi-step ETL that transforms raw event logs into the canonical datasets the company reports on, shipping 18 new metric definitions with 12 months of rebuilt history and owning the documentation so teams read the same number the same way across 19 markets.
- Set the data integrity standards and delivery SLAs for 30+ production pipelines at 20 TB daily, and built PulseOps to enforce them: automated validation across every job, freshness and quality checks, and alerting before a stakeholder opens a stale dashboard. Wrote the standards doc the team adopted and deployed it across 5 regions.
- Build foundational data products and reporting that let analysts and business teams self-serve rather than queue behind a data engineer, and translate ambiguous stakeholder questions into the data models and technical requirements that answer them.
- Caught a revenue calculation error during validation that would have published wrong numbers to every user dashboard at launch, on a rebuild already committed to a date. Shipped on time and correct rather than choosing between them.
- Went outside the job description repeatedly: diagnosed a stalled ranking feature others were fixing by tuning the query, found the constraint was the access pattern, and moved computation upstream into Flink over a 1.5B+ record stream, cutting query time and compute cost by orders of magnitude.
- Support machine learning work end to end, building the feature and evaluation datasets a production model consumed and the serving layer around its deployment. Mentor junior engineers on modeling and code review.

Delta Air Lines Inc. | Data Engineer

Atlanta, GA | Aug 2021 to May 2025

- Modeled the loyalty domain into Teradata for 500K+ customers and built the segmentation reporting and dashboards marketing and operations teams ran campaigns from, working directly with non-technical stakeholders as requirements shifted.
- Led migration of 50+ legacy Informatica workflows onto AWS Glue PySpark and Lambda with GitHub-based CI/CD, cutting licensing costs 65%, roughly \$850K a year.
- Designed and operated production Airflow pipelines integrating vendor APIs and legacy systems into S3 and Teradata, cutting ETL runtime 65% from 4.5 hours to 1.5 and holding 99.9% accuracy with idempotent DAGs and reconciliation checks.
- Built asynchronous event-driven ingestion on Kinesis, Lambda, and SNS carrying 3M+ daily transactions at 99.9% uptime, cutting end-to-end latency from 2 hours to under 5 minutes.
- Built a serverless S3 data lake, moving 15 TB with AWS DMS and cataloging it through Glue Catalog and Lake Formation with the access controls required for GDPR and CCPA.

TECHNICAL SKILLS

Languages: Python (pandas, PySpark), SQL, Scala, Bash

Transformation & Modeling: dbt, multi-step ETL, canonical datasets, dimensional modeling, semantic and metrics layers, schema design

Pipelines & Orchestration: Airflow, Spark, Flink, Kafka, Kinesis, Hive, Trino/Presto; GitHub version control, CI/CD

Warehouse & Storage: Snowflake, BigQuery, Apache Iceberg on S3, ClickHouse, Teradata, PostgreSQL, MySQL

Reporting & Self-Serve: Hex, Tableau, Power BI, Streamlit; dashboards, metric definitions, self-serve data products

Integrity & Reliability: Data SLAs, automated validation, freshness monitoring, alerting, reconciliation, lineage, documentation

EDUCATION

Rice University, MS, Space Engineering (concentration in Data Science)

Houston, TX | 2019 to 2021

Amrita Vishwa Vidyapeetham, BE, Aerospace Engineering

Coimbatore, India | 2015 to 2019

Graduate work: exploratory analysis and anomaly detection on large scientific datasets in Python.